

Case Study 4.2: Low carb lunch

V.1.3



This paper is about putting the **settings** I had (about 2 years ago) established **for my entire usual meal spectrum** (~50 – 120 g carbs/meal) to the test, with two extremely low-carb lunches.

Method

FCL (no carb inputs, no user boli) with dev variant of AAPS 3.2.0.4 w/autoISF 3.0.1:
Lyumjev 100 (DIA 7h/*remark added later: 8.5 h slightly better*) in Combo pump w/ 10mm Teflon cannula (0-48h)
2 x G6 overlapping (see [case study 1.5](#); sensors used ~ d3 – d15; xDrip, no smoothing in AAPS)
TDD ~ 35 U; profile basal ~ 14 U (0.41...0.75 U/h); profile_ISF 36...44 mg/dl/U; iobTH% = 0.6

Key settings*) for entire meal spectrum (~ 20 ... 90 g carb per meal):

SMB size limited at ~ 3.5 U (=2.9 x 120 minutes basal)

autoISFmax = 2.9; SMB delivery ratio = 0.75 fixed

bgAccel_ISF_weight = 0.22; break_weight 0.12; lower_ISF-range_weight 0.7; higher_ISF-range_weight 0.1; pp_ISF_weight = 0.03; dura_ISF_weight 0.8

*) **Caution: Do not copy settings** from others,
not even for starting your tuning.

Why, see FCL e-book [section 4.1](#).

Fish + veggie lunches, managed by autoISF FCL

Settings that were better proven to work for fairly high carb meals (see e.g. Xmas [case study 4.3](#)) were put to the test with two (for “my spectrum”) extremely low carb meals.

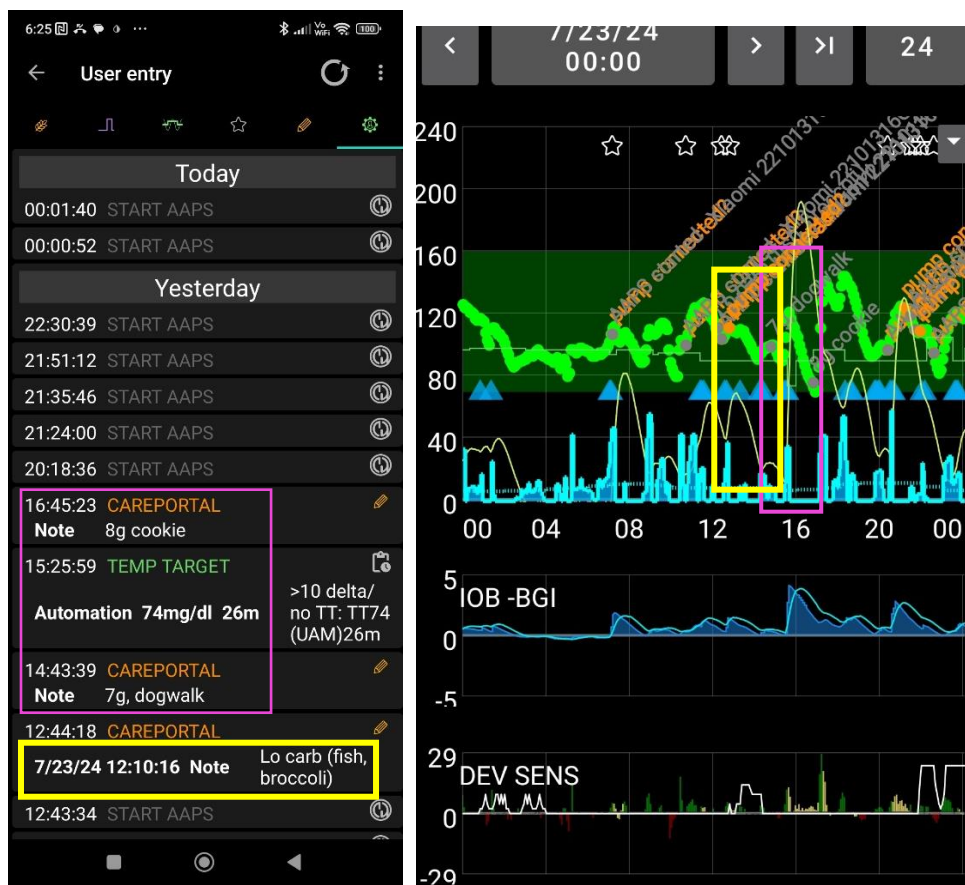
Two very similar meals were set up for two consecutive days, with an option to lower the iobTH, or any of the ..._ISF_weights, then.

As the FCL e-book suggests (in [section 5.2.2](#)), it would be easy to “nudge” my FCL for an “outlier” of comparatively low insulin need

- by lowering iobTH% either directly, or indirectly (via low %profile, high TT + exercise button)
- by elevating the effective ISF via lower bgAccel_ISF_weight and / or pp_ISF_weight, and/or via a lower % profile



38 The first example shown was a cod fillet plus broccoli with herbal crème cheese (12:10)



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40 As that went well without any manual intervention...

41 other than, 2.5 -3 hours after the meal, 2 small cookies before/after a dogwalk

42 ...no settings changes were made for the second meal, which had about same carbs and
43 calories.

44 Note that iob stayed under 4U all the time, and under 2U in first 2 hours after meal start. That
45 is well below my iobTH of 6 U.

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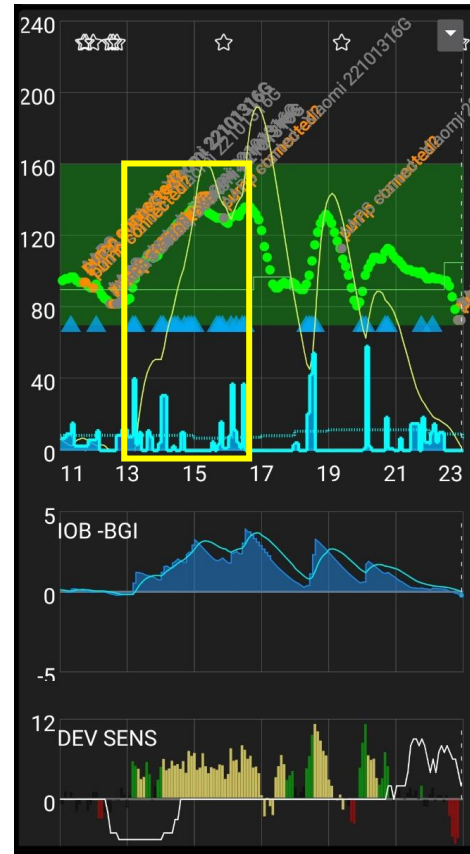
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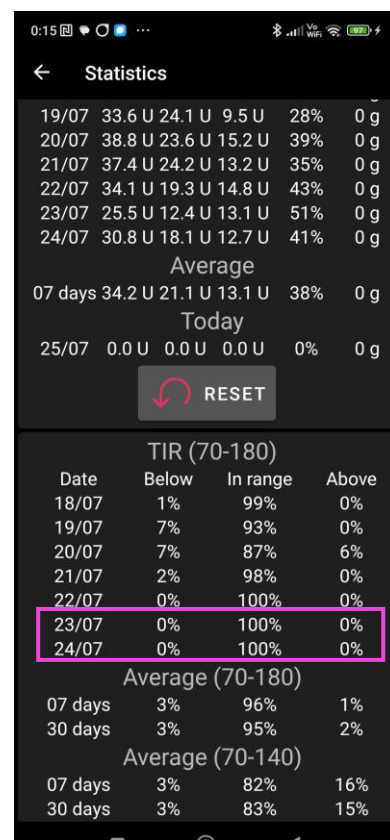
The 2nd meal, lunch on another day, was
2 small tuna steaks, and black beans (12:50).

Now the initial iob rose stronger (probably
because black beans have more carb than
broccoli) but remained in the 3 U magnitude
in first 2 hours after meal start



Despite aggressive settings that *could* quickly
deliver big SMBs, and exceed iobTH, we see very
moderate SMBs, and no hypo tendency at all.

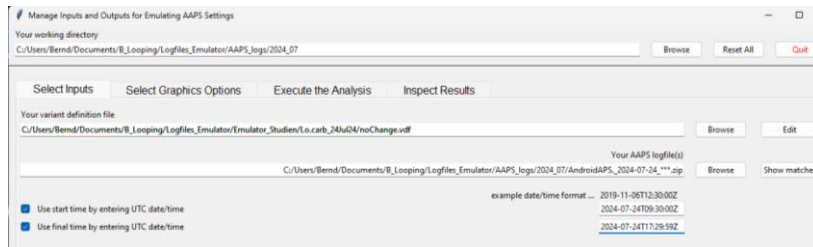
On both days, 100% TIR was achieved:



94 Analysis using the Emulator

95 Let us have a look how the autoISF factors contributed to provide appropriate iob (well
96 below iobTH in this low carb case).

97 Following [section 10.2.1 – 10.2.4](#) of the FCL e-book, analysis of the 2nd meal (24th July):



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99 Execute analysis / run emulation / inspect results / csv =>

100 In column C: AAPS time = UTC time (B) + 2/24 ; format without seconds

101 Hide not important lines and columns =>

	A	C	E	F	G	L	M	N	Q	R	S	V	W	X	Y	AA	AB	AC	AD	AE	AF	AH	AI	AK	AO	AQ	AR
1	2		AAPS	bg	ac	brake	targ		eff.	tol.	ISF	dura	min	para	para	para	para	if	pp	delta	dura	final	ISF	ISF	ISF	ISF	ISF
3	id	time				iob			emul	emul	orig	utes	avg	correl	durat	last	next	emul	emul	emul	emul	prof	emul	emul	emul	emul	emul
10	6	11:51	92	92	90	0,02	6	7,8	0,89	75	96,1	1	45	-1,3	-1,5	0,97	0,89	1	1	1	0,89	44	49,5	0	0	0,39	
11	7	11:56	91	91	90	0	6	7,8	0,87	80	95,8	1	25	-1,3	-1,6	0,97	0,87	1	1	1	0,87	44	50,3	0	0	0,385	
12	8	12:06	93	90	-0,03	4,92	6,4	0,74	90	95,6	0	0	0	0	0	0	1	0,9	1	1	1	0,74	43	56,1	0,14	0,1	0,731
13	9	12:11	90	90	0,08	3,84	4,99	0,55	90	95,1	0	0	0	0	0	0	1	0,86	1	1	1	0,55	43	78,1	0	0	0
14	10	12:16	87	90	0,03	3,84	4,99	0,54	5	88,5	0	0	0	0	0	0	1	0,84	1	1	1	0,54	43	80,1	0	0	0
15	11	12:21	85	90	-0,02	3,84	4,99	0,53	10	87,3	0	0	0	0	0	0	1	0,82	1	1	1	0,53	43	81,4	0	0	0
16	12	12:26	84	84	90	-0,07	4,92	6,4	0,67	10	85,3	0,9995	20	-1,2	-0,5	0,92	0,82	1	1	1	0,67	43	64,1	0	0	0	
t ->	12:31	82	82	90	-0,11	6	7,8	0,8	15	84,5	0,9973	25	-1,2	-0,8	0,95	0,8	1	1	1	1	0,8	43	53,5	0	0	0	
	12:51	84	90	-0,19	6	7,8	0,87	35	83,5	0,9929	45	1,1	1,7	1,06	0,82	1	1	1	1	1	0,87	43	49,5	0,1	0	0,75	
	12:56	85	90	-0,17	6	7,8	0,87	40	83,7	0,9976	25	1,5	2	1,05	0,82	1	1	1	1	1	0,87	43	49,5	0,1	0	0,77	
	13:01	84	90	-0,14	6	7,8	0,82	45	83,7	0,8103	45	1,4	1,7	1	0,82	1	1	1	1	1	0,82	41	50,1	0	0	0,47	
20	16	13:06	85	85	90	-0,14	6	7,8	0,83	50	83,8	0,9998	30	0,5	0,4	0,99	0,82	1	1	1	0,82	41	49,7	0,02	0	0,495	
21	18	13:11	91	90	-0,14	6	7,8	1,15	0	91	0,9826	20	4	5,7	1,31	0,87	1	1	1	1	1,15	41	35,8	1,01	0,7	2,57	
23	19	13:16	96	90	0,74	6	7,8	1,3	0	96	1	25	5,7	7,4	1,38	0,94	1	1	1	1	1,3	41	31,6	0,79	0,5	0,22	
24	20	13:21	100	90	1,2	6	7,8	1,18	5	98	1	40	5	5,8	1,18	1	1,12	1	1	1	1,18	41	34,8	0,06	0	0,67	
25	21	13:26	102	102	90	1,19	6	7,8	0,9	10	99,3	0,9997	20	2,4	1,1	0,85	1	1,06	1	1,01	0,9	41	45,7	0	0	0	
26	22	13:31	104	104	90	1,12	6	7,8	0,92	10	102	0,9997	25	1,5	0,3	0,86	1	1,06	1	1,02	0,92	41	44,8	0	0	0	
27	23	13:36	108	108	90	1,03	6	7,8	1,08	15	104	0,9992	35	2,6	2,3	0,96	1	1,12	1	1,03	1,08	41	38	0,08	0	0,71	
28	24	13:41	110	110	90	1,01	6	7,8	1,02	15	106	1	40	2,3	2	0,96	1	1,06	1	1,04	1,02	41	40,2	0	0	0,51	
t ->	13:46	108	108	90	0,96	6,36	8,27	1,02	15	108	0,9876	40	0,5	-0,3	0,93	1	1	1	1	1,04	1,02	41	40,2	0	0	0	
	13:51	108	108	90	0,88	6,36	8,27	1,02	20	108	0,9886	45	-0,4	-1,2	0,91	1	1	1	1	1,05	1,02	41	40,3	0	0	0	
	13:56	109	109	90	0,77	6,36	8,27	1,06	25	108	0,9853	45	-0,6	-1,3	0,93	1	1,03	1	1	1,07	1,06	41	38,8	0	0	0,29	
	14:01	112	90	0,7	6,36	8,27	1,43	30	108	0,9999	20	2,9	4,4	1,35	1	1,09	1	1	1,08	1,43	40	28,1	0,71	0,5	2,003		
33	29	14:06	116	90	1,27	6,36	8,27	1,39	30	110	1	20	4,1	5,6	1,31	1,01	1,12	1	1,09	1,39	40	28,7	0,42	0,3	1,98		
34	30	14:11	118	90	1,64	6,36	8,27	1,14	10	115	0,9974	40	2,6	3	1,07	1,01	1,06	1	1,04	1,14	40	35,2	0	0	0,023		
31	31	14:16	120	120	90	1,53	6,36	8,27	1,05	10	118	1	20	1,9	1,4	0,93	1,01	1,06	1	1,04	1,05	40	38,2	-0,03	0	0,4	
32	32	14:21	123	90	1,41	6,36	8,27	1,16	15	119	1	30	2,7	2,7	1	1,02	1,09	1	1,07	1,16	40	34,6	0,66	0,4	0,16		
33	33	14:26	125	125	90	1,71	6,36	8,27	1,13	20	120	0,9999	25	2,1	1,9	0,97	1,02	1,06	1	1,09	1,13	40	35,5	0,34	0,2	0	
34	34	14:31	127	127	90	1,78	6,36	8,27	1,14	20	123	1	30	2	1,8	0,98	1,02	1,06	1	1,1	1,14	40	35,2	0,48	0,3	0	
39	35	14:36	128	128	90	1,96	6,36	8,27	1,09	20	125	0,9999	20	1,1	0,6	0,93	1,02	1,03	1	1,1	1,09	40	36,7	0,03	0	0,643	
40	36	14:41	130	130	90	1,87	6,36	8,27	1,15	20	127	1	35	1,6	1,4	0,98	1,02	1,06	1	1,11	1,15	40	34,7	0,89	0,6	0	

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		A	C	E	F	G	L	M	N	Q	R	S	V	W	X	Y	AA	AB	AC	AD	AE	AF	AH	AI	AK	AO	AQ	AR
->	14	44	14:46	130	130	90	2,35	6,36	8,27	1,15	25	127	0,9981	30	0,7	0,3	0,95	1,02	1	1	1,14	1,15	40	34,8	0	0	0	
	14	45	14:51	133	133	90	2,21	6	7,8	1,15	30	128	0,9931	45	1,2	1	0,98	1,03	1,09	1	1,17	1,15	40	34,8	0,52	0,3	0	
	14	45	14:56	135	90	2,36	6	7,8	1,18	30	130	0,9999	20	2,3	2,8	1,09	1,03	1,06	1	1,18	1,18	40	34	0,91	0,6	0		
44	40	15:01	137	90	2,79	6	7,8	1,21	35	131	1	35	2,1	2,3	1,03	1,03	1,06	1	1,21	1,21	40	33	0,67	0,5	0			
45	41	15:06	136	136	90	3,12	6	7,8	1,09	35	132	0,9796	20	-0,1	-1,3	0,89	1,03	1	1	1,22	1,09	40	36,9	0	0	0		
46	42	15:11	135	135	90	2,94	6	7,8	1,08	40	132	1	25	-1,3	-2,4	0,86	1,03	1	1	1,25	1,08	40	37,1	0	0	0		
47	43	15:16	136	136	90	2,76	6	7,8	1,24	45	133	0,9991	45	0	-0,2	0,97	1,03	1,03	1	1,28	1,24	40	32,2	0	0	0		
48	44	15:26	133	133	90	2,39	6,36	8,27	1,35	55	133	0,9983	35	-1,5	-2	0,94	1,03	1	1	1,35	1,35	40	29,7	0	0	0		
49	45	15:30	132	132	90	2,21	6,36	8,27	1,43	60	133	0,9994	30	-1,3	-1,5	0,97	1,03	1	1	1,38	1,43	40	28	0	0	0,337		
50	46	15:36	131	90	2,05	6,36	8,27	1,5	65	133	0,9995	20	-1,1	-0,9	1,03	1,03	1	1	1,41	1,5	40	26,7	-0,04	0	0	0		
51	47	15:41	130	90	1,88	6,36	8,27	1,53	70	133	1	25	-0,9	-0,8	1,03	1,02	1	1	1,44	1,53	40	26,2	0,11	0	0,697			
->	52	48	15:46	130	90	1,82	6,36	8,27	1,56	75	132	0,9985	25	-0,3	0,1	1,08	1,02	1	1	1,47	1,56	40	25,7	0,27	0,2	1,017		
	53	49	15:51	130	90	1,94	6,36	8,27	1,59	80	132	1	30	0,1	0,4	1,08	1,02	1	1	1,5	1,59	40	25,2	0,28	0,2	0,08		
	54	50	16:01	127	127	90	1,83	6	7,8	1,5	90	132	1	25	-1,3	-1,6	0,97	1,02	1	1	1,56	1,5	39	26	-0,04	0	0,46	
55	51	16:06	130	90	1,71	6	7,8	1,59	95	132	0,8717	45	0,2	0,4	1	1,02	1,09	1	1,59	1,59	39	24,6	0,93	0,6	2,34			
56	52	16:11	131	90	2,45	6	7,8	1,62	100	132	0,9959	25	1,9	2,8	1,13	1,03	1,03	1	1,62	1,62	39	24,1	0,12	0	0,72			
57	53	16:16	132	90	2,35	6	7,8	1,65	105	132	1	40	1,4	1,8	1,09	1,03	1,03	1	1,65	1,65	39	23,7	0,34	0,2	0,29			
58	54	16:21	131	131	90	2,4	6	7,8	1,46	110	132	0,9949	20	-0,7	-1,9	0,87	1,03	1	1	1,68	1,46	39	26,7	0	0	0,1		
59	55	16:26	134	90	2,24	6	7,8	1,71	115	132	0,9998	40	1,7	2	1,07	1,03	1,09	1	1,71	1,71	39	22,8	1,14	0,8	2,76			
60	56	16:31	136	90	3,07	6	7,8	1,74	120	132	1	45	2	2,4	1,07	1,03	1,06	1	1,74	1,74	39	22,4	1,29	0,9	4,9	0	0	
61	57	16:36	136	136	90	3,78	6	7,8	1,77	125	132	0,9971	35	1,1	1,1	1	1	1,03	1	1,78	1,77	39	22	0	0	0		
->	62	58	16:41	136	136	90	3,58	6	7,8	1,59	130	132	0,9896	20	-0,5	-1,7	0,88	1,03	1	1	1,81	1,59	39	24,5	-0,41	0	0	
	63	59	16:51	132	132	90	3,15	6,36	8,27	1,5	140	132	0,9996	20	-2,3	-3,1	0,9	1,02	1	1	1,68	1,5	39	25,9	0	0	0	
	64	60	16:56	129	129	90	2,93	6,36	8,27	1,53	145	132	1	25	-3	-3,9	0,9	1,01	1	1	1,7	1,53	39	25,5	0	0	0	
65	61	17:01	123	123	90	2,72	6,36	8,27	0,9	15	130	0,9984	20	-5	-6,3	0,85	1,01	1	1	1,07	0,9	38	42	0	0	0		
66	62	17:06	114	114	90	2,5	6,36	8,27	0,71	0	114	0,9998	20	-8,5	-11	0,71	1	1	1	1	0,71	38	53,6	0	0	0		
67	63	17:11	108	108	90	2,29	6,36	8,27	0,85	0	108	0,9997	45	-7,9	-9,2	0,85	1	1	1	1	0,85	38	44,7	0	0	0		
68	64	17:16	101	90	2,08	6,36	8,27	0,94	0	101	1	20	-6,9	-6,7	1,03	0,92	1	1	1	0,94	38	40,2	0	0	0	0		
69	65	17:21	93	93	90	1,89	6,36	8,27	0,83	0	93	0,9999	25	-7,3	-7,4	0,98	0,83	1	1	1	0,83	38	45,7	0	0	0		
70	66	17:26	91	91	90	1,69	6,36	8,27	0,82	5	92	0,9966	25	-4,6	-3,7	0,89	0,82	1	1	1	0,82	38	46,5	0	0	0		
71	67	17:31	91	90	1,52	6,36	8,27	0,96	10	91,7	0,9997	20	-0,1	2,7	1,17	0,82	1	1	1	0,96	38	39,7	0	0	0	13		
72	68	82									0,53					0,71	0,8	1	1	1	36	22	0	0	0	to		
93	94	137									1,77					1,57	1,03	1,27	1	1,81		44	81,4		1,1	16		
99																									10,8	2,82		

Discussion of results

Optimization potential for low carb meals?

The investigated low carb meals were managed very well with the settings that also suit higher carb meals.

In case there were ideas for potential further improvements:

- The difference that modified settings *would make* could be analyzed using the Emulator with a “what-if” vdf. See FCL e-book [section 10.3](#) for analysis on your PC, and (especially neat for AAPS:) the *real-time* emulator analysis of a *what-if* question, with speech synthesis on your smartphone, telling you for every loop decision how it would be impacted. See FCL e-book [section 11.4](#)
- It must be critically checked whether, after such optimization, *other* meals in your spectrum then might suffer. **It is problematic, to fine tune just for one kind of meal.** (See FCL e-book [section 8](#), and [case study 8.2](#)),
- However, in case of fairly consistent different eating habits at lunch vs at dinner (for instance high carb lunches, low carb dinners): The user then could choose to create two differentiated sets of autoISF related parameters, suited to each rough time slot, and optimize both independently. (This would come on top of probably already using different profileISFs at lunch vs at dinner time).
More see in [sections 5.1.4 and 5.1.5](#).

Was I just lucky – Could there be trickier cases?

We know that **super big meals** do not pose much extra difficulty, if the system was set up for a diet spectrum that included meals with max carb “burn rate” of around 30g/h (ref: Chapter 8 in: <https://github.com/danamlewis/artificialpancreasbook/> -)

The presented case study showed that **low carb meals** are also easily managed.

However: **Reduced-size meals (or high carb snacks)** that trigger really aggressive first SMBs due to (a limited amount of) fast resorbing carbs **would be more problematic**, and might require a manual intervention to “tame” the loop (...or another late snack, to prevent going low).

More regarding how to master **snacks** in FCL, see [section 5.2.2.3](#), Snack example, and [case study 5.2](#).